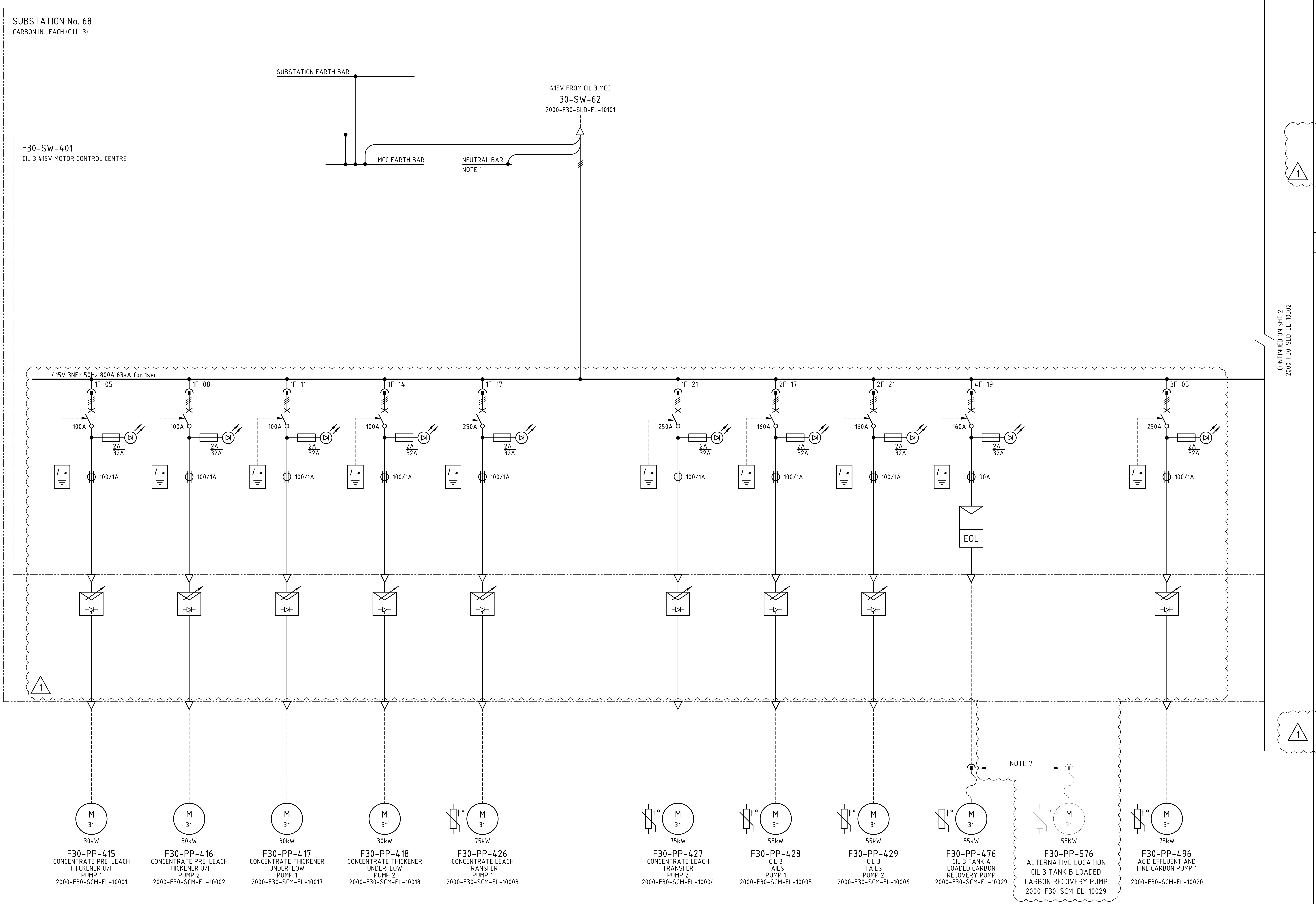
SUBSTATION No. 68
CARBON IN LEACH (C.I.L. 3)
SUBSTATION EARTH BAR
415V FROM CIL 3 MCC
30-SW-62
2000-F30-SLD-EL-10101
F30-SW-401
CIL 3 415V MOTOR CONTROL CENTRE
MCC EARTH BAR
NEUTRAL BAR
NOTE 1
CONTINUED ON SHT 2
2000-F30-SLD-EL-10302
415V 3NE~ 50Hz 800A 63kA for 1sec
1F-05
1F-08
1F-11
1F-14
1F-17
1F-21
2F-17
2F-21
4F-19
3F-05
100A
250A
160A
2A
32A
100/1A
90A
EOL
1
NOTE 7
M
3~
30kW
F30-PP-415
CONCENTRATE PRE-LEACH
THICKENER U/F
PUMP 1
2000-F30-SCM-EL-10001
30kW
F30-PP-416
CONCENTRATE PRE-LEACH
THICKENER U/F
PUMP 2
2000-F30-SCM-EL-10002
30kW
F30-PP-417
CONCENTRATE THICKENER
UNDERFLOW
PUMP 1
2000-F30-SCM-EL-10017
30kW
F30-PP-418
CONCENTRATE THICKENER
UNDERFLOW
PUMP 2
2000-F30-SCM-EL-10018
t°
75kW
F30-PP-426
CONCENTRATE LEACH
TRANSFER
PUMP 1
2000-F30-SCM-EL-10003
75kW
F30-PP-427
CONCENTRATE LEACH
TRANSFER
PUMP 2
2000-F30-SCM-EL-10004
55kW
F30-PP-428
CIL 3
TAILS
PUMP 1
2000-F30-SCM-EL-10005
55kW
F30-PP-429
CIL 3
TAILS
PUMP 2
2000-F30-SCM-EL-10006
55kW
F30-PP-476
CIL 3 TANK A
LOADED CARBON
RECOVERY PUMP
2000-F30-SCM-EL-10029
55KW
F30-PP-576
ALTERNATIVE LOCATION
CIL 3 TANK B LOADED
CARBON RECOVERY PUMP
2000-F30-SCM-EL-10029
75kW
F30-PP-496
ACID EFFLUENT AND
FINE CARBON PUMP 1
2000-F30-SCM-EL-10020

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM 30-SW-62 INCOMER. SIGNAGE SHALL BE PLACED ON F30-SW-401 INCOMING SUPPLY SECTION READING "M.E.N. LINK LOCATED IN 30-SW-62".
2. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
3. ALL DRIVE I/O VIA COMMS NETWORK TO DCS.
4. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
5. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
6. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
7. SECOND DECONTACTOR INSTALLED ABOVE CIL3 TANK B, AT THE F30-PP-476 ALTERNATE LOCATION. FLEX CABLE IS INSTALLED BETWEEN THE CIL3 TANK A AND CIL3 TANK B DECONTACTORS, WITH A PLUG LOCATED AT THE CIL3 TANK A DECONTACTOR END. CONNECT THE PLUG TO THE CIL3 TANK A DECONTACTOR ONLY WHEN F30-PP-476 IS LOCATED AT CIL3 TANK B.

1

LEGEND

	DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 2, 3
	DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 2, 3
EOL	ELECTRONIC OVERLOAD UNIT NOTE 2, 3
	THERMAL OVERLOAD UNIT NOTE 4
	VARIABLE SPEED DRIVE NOTE 2, 3
/ >	EARTH LEAKAGE RELAY NOTE 2, 3
	FIELD-MOUNTED DECONTACTOR
t°	MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
RTD	MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
	MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
(49) (63) (71)	TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
	CIRCUIT BREAKER (NOTE 6)
xxxA	ST TRIP SET POINT
xxxA	CB SIZE
AFD	ARC FLASH DETECTION
ECR	EARTH CONTINUITY RELAY
1 E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
	63A 3 PHASE SOCKET OUTLET

NORTHERN STAR RESOURCES LIMITED

Approved	☒
Approved as Noted	☐
Not Approved	☐
For Information Only	☐
Signed [redacted]	Date 05/12/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

DRG No.	REFERENCES
2000-F30-SLD-EL-10302	SUBSTATION NO. 68 MCC F30-SW-401 SINGLE LINE DIAGRAM SHEET 2
2000-F30-SLD-EL-10101	SUBSTATION NO. 68 MCC F30-SW-62 SINGLE LINE DIAGRAM SHEET 1
2000-F30-DRG-EL-10001	SUBSTATION NO. 68 MCC AND GENERAL EQUIPMENT LAYOUT
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	28OCT25	FAT REDLINES INCORPORATED	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
0	17MAR25	APPROVED FOR CONSTRUCTION	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

NORTHERN STAR
RESOURCES LIMITED

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: FIMISTON PLANT - CARBON IN LEACH (CIL 3)
SUBSTATION NO. 68 MCC F30-SW-401
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F30-SLD-EL-10301	1